

The Antarctic Baseline: Quantifying Environmental Dynamics in Near Zero-Radiance Regimes

Part 2: Ensemble Learning and Explainable Machine Learning for Antarctic Surface Albedo Prediction

Recap from Part 1

Part 1 of this study established a statistical baseline for modeling Antarctic surface albedo over the Larsen C Ice Shelf using a custom Degree-2 Polynomial Regression model optimized through Gradient Descent. Using ten years of satellite-derived environmental observations (2015-2025), the analysis demonstrated that a lightweight regression framework could successfully capture the dominant reflectance behavior of the ice shelf.

However, residual diagnostics revealed that model performance declined under rare low-albedo conditions. While the polynomial model represented average environmental behavior effectively, it struggled to capture abrupt reflectance reductions associated with less common surface states. These findings suggest that a single global regression surface may not be flexible enough to describe the full range of environmental variability present within the Antarctic system.

Part 2 builds directly on this foundation by introducing ensemble-based machine learning approaches that can learn localized non-linear relationships without assuming a single global functional form.

Problem Statement

This notebook investigates whether tree-based ensemble learning methods can improve the prediction of Antarctic surface albedo compared to the polynomial baseline developed in Part 1. By evaluating Decision Trees, Random Forests, AdaBoost, and explainable machine learning techniques, the study examines whether localized non-linear models can better represent complex reflectance behavior across the Larsen C Ice Shelf.

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv("C:/Users/User/Downloads/larsen_c_cleaned_data.csv")
df.head()
```

Out[1]:

	date	surface_temp	solar_zenith	surface_albedo
0	2015-01-01	-4.490099	53.772216	0.733728
1	2015-01-02	-3.446475	54.129804	0.732435
2	2015-01-03	-3.397931	51.971582	0.731124
3	2015-01-04	-5.659448	57.299931	0.733065
4	2015-01-05	-19.234894	56.139842	0.732165

The dataset used in this notebook was generated during Part 1 of the study and contains satellite-derived environmental observations extracted over the Larsen C Ice Shelf. The cleaned dataset includes surface temperature, solar zenith angle, and surface albedo measurements that were prepared for machine learning analysis.

```
In [11]: df.describe()
```

Out[11]:

	surface_temp	solar_zenith	surface_albedo
count	2699.000000	2699.000000	2699.000000
mean	-21.292886	68.316447	0.708256
std	9.916246	10.095973	0.049334
min	-59.593810	51.186503	0.380821
25%	-28.328271	59.039289	0.675090
50%	-20.285292	67.161757	0.719094
75%	-13.499354	78.139082	0.745672
max	0.091557	94.162618	0.808750

Surface temperature and solar zenith angle are used as predictor variables, while surface albedo serves as the target variable to be estimated by the ensemble learning models.

```
In [3]: X=df[['surface_temp', 'solar_zenith']]
y=df['surface_albedo']
```

Decision Tree Regression

In Part 1, a single polynomial equation was used to describe the relationship between environmental variables and surface albedo. While this approach successfully captured the dominant reflectance behavior across the Larsen C Ice Shelf, the diagnostic analyses showed that it struggled to reproduce rare low-albedo events and tended to smooth extreme observations toward the overall dataset mean.

A Decision Tree takes a different approach. Instead of fitting one global equation, it repeatedly divides the dataset into smaller groups based on feature values and learns local prediction rules within each region. This allows the model to capture non-linear relationships and sudden changes that may be difficult to represent using a single regression.

For Antarctic surface conditions, this flexibility may help improve predictions during unusual reflectance events where albedo departs from its dominant ice-covered state.

```
In [6]: from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error, r2_score

X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2,random_s
tree=DecisionTreeRegressor(max_depth=4,random_state=726)
tree.fit(X_train, y_train)
pred_tree=tree.predict(X_test)
mse_tree=mean_squared_error(y_test, pred_tree)
r2_tree=r2_score(y_test, pred_tree)
print("Decision Tree MSE :", mse_tree)
print("Decision Tree R^2 :", r2_tree)
```

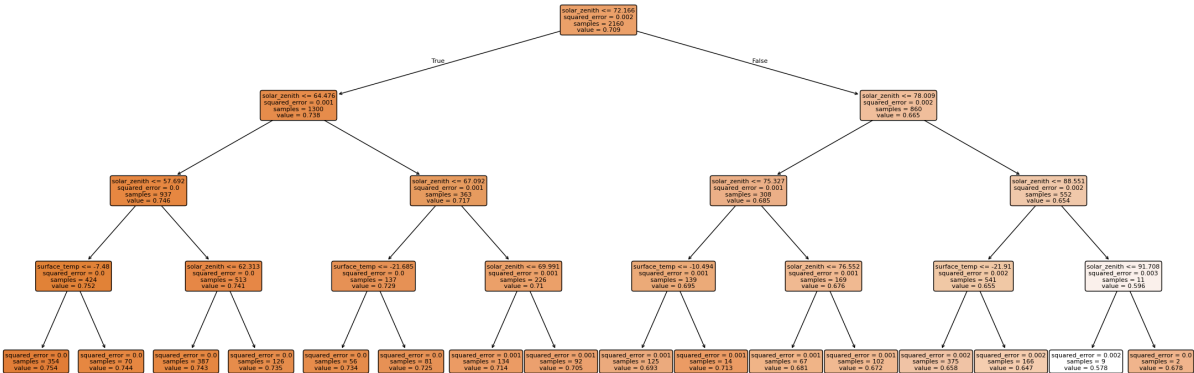
Decision Tree MSE : 0.0008241524737932535
Decision Tree R^2 : 0.658254325180921

Interpretation

The Decision Tree achieved an approximate Mean Squared Error (MSE) of 0.0008241524737932535 and an R^2 score of 0.658254325180921 on the test dataset.

- Compared to the polynomial regression baseline from Part 1, the Decision Tree produced a slightly lower prediction error. Although the improvement in MSE is modest, the tree introduces a fundamentally different learning strategy by dividing the data into localized decision regions rather than fitting a single global equation. This allows the model to better represent non-linear environmental behavior.
- An R^2 value of approximately 0.66 indicates that the model explains about 66% of the observed variation in surface albedo using only surface temperature and solar zenith angle. This suggests that a substantial portion of Antarctic reflectance variability can be captured using a small set of environmental predictors.

```
In [9]: from sklearn.tree import plot_tree
plt.figure(figsize=(28,10))
plot_tree(tree,feature_names=['surface_temp', 'solar_zenith'],filled=True,round
plt.show()
```



Interpretation of the Decision Tree

The structure of the Decision Tree provides insight into how the model organizes environmental conditions before making albedo predictions.

1. Solar Zenith

The first split of the tree occurs on solar zenith angle, and the upper levels continue to rely heavily on this variable. This suggests that large-scale differences in Antarctic surface albedo are primarily associated with changes in solar illumination geometry rather than temperature alone.

In practical terms, the amount and angle of incoming sunlight appear to explain most of the broad reflectance patterns observed across the Larsen C Ice Shelf. The tree therefore uses solar zenith angle to create the major environmental groups before considering any additional factors.

2. Temperature

Temperature first appears deeper in the tree, where it is used to refine predictions within already-defined solar conditions. This does not mean temperature is of lesser importance. Instead, it suggests that temperature acts as a more localized influence on albedo after the broader illumination regime has been established.

The model therefore treats solar geometry as the primary source of variation and temperature as a secondary factor that helps explain smaller differences within those groups.

Capturing Extreme Albedo Conditions

One of the main limitations of the Polynomial Regression model in Part 1 was its tendency to smooth predictions toward the average behavior of the dataset. As a result, very low-albedo observations were often predicted as higher values.

The Decision Tree overcomes this limitation. Instead of forcing all observations to follow a rigid mathematical rule, the tree can isolate rare environmental conditions and assign them their own prediction region.

The branch leading to the low-albedo leaf value of *0.578* demonstrates this behavior. By repeatedly applying solar zenith thresholds, the model successfully separates a small group of extreme observations from the rest of the dataset. This shows how tree-based methods can capture unusual environmental states that are difficult to represent using a single global regression equation.

Overall, the Decision Tree provides early evidence that localized non-linear models are better suited for representing complex Antarctic albedo behavior than a single polynomial relationship.

Random Forest Regression

A Decision Tree learns a single set of splitting rules from the training data. While it can capture non-linear relationships, its predictions may depend heavily on the specific samples used during training. Random Forest addresses this limitation by building many Decision Trees on different subsets of the data and combining their predictions. Instead of relying on one tree, the final prediction is obtained by averaging the outputs of multiple trees. This approach typically reduces prediction variance, improves generalization, and produces more stable results when environmental conditions vary across the dataset.

```
In [10]: from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score
rf=RandomForestRegressor(n_estimators=100,random_state=726)
rf.fit(X_train, y_train)
pred_rf=rf.predict(X_test)
mse_rf=mean_squared_error(y_test,pred_rf)
r2_rf=r2_score(y_test, pred_rf)
print("Random Forest MSE :", mse_rf)
print("Random Forest R^2  :", r2_rf)
```

```
Random Forest MSE : 0.0009256062106746325
Random Forest R^2  : 0.6161851973484638
```

Interpretation

The Random Forest model achieved a Mean Squared Error (MSE) of *0.0009256062106746325* and an R^2 score of *0.6161852* on the test dataset. Contrary to expectations, the ensemble did not improve upon the baseline Decision Tree. Both the MSE increased and the proportion of explained variance decreased compared to the decision tree.

This suggests that the relationships present within the dataset may already be captured effectively by a relatively simple decision structure. By averaging the predictions of many trees, the Random Forest produced smoother predictions, but this additional averaging reduced its ability to represent some of the localized patterns identified by the baseline Decision Tree.

An R^2 value of approximately *0.616185* indicates that the model explains around 62% of the observed variation in surface albedo. While this still represents a meaningful relationship between the environmental variables and surface reflectance, the performance is lower than the Decision Tree benchmark established earlier.

Random Forest can estimate how much each predictor contributes to the model's decisions. This helps identify which environmental variables have the strongest influence on Antarctic surface albedo predictions.

```
In [11]: imp=pd.DataFrame({
    "Variables": X.columns,
    "Importance": rf.feature_importances_})
imp=imp.sort_values("Importance", ascending=False)
print(imp)
```

	Variables	Importance
1	solar_zenith	0.826902
0	surface_temp	0.173098

Variable Importance Results

The Random Forest model identified solar zenith angle as the dominant variable, accounting for approximately 82.7% of the total predictive importance. Surface temperature contributed the remaining 17.3%. This result is consistent with the Decision Tree analysis, where solar zenith angle controlled the upper levels of the tree structure and determined the major environmental groupings before temperature was considered. The consistency across multiple models suggests that changes in solar illumination geometry mainly influence the large-scale albedo variability across the Larsen C Ice Shelf, while temperature acts mainly as a secondary local adjustment factor.

AdaBoost Regression

Unlike Random Forest, which averages many independent trees, AdaBoost builds trees sequentially. Each new tree pays greater attention to observations that previous trees predicted poorly. The Polynomial Regression model in Part 1 struggled with rare low-albedo observations, while the Decision Tree showed that localized rules could better represent these conditions. AdaBoost allows us to test whether focusing on difficult observations can further improve predictive performance across the Larsen C Ice Shelf.

```
In [13]: from sklearn.ensemble import AdaBoostRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error, r2_score
ada=AdaBoostRegressor(estimator=DecisionTreeRegressor(max_depth=3),n_estimators=100)
ada.fit(X_train, y_train)
pred_ada=ada.predict(X_test)
mse_ada=mean_squared_error(y_test, pred_ada)
r2_ada=r2_score(y_test, pred_ada)
print("AdaBoost MSE :", mse_ada)
print("AdaBoost R^2 :", r2_ada)
```

AdaBoost MSE : 0.0008440754890491493
AdaBoost R^2 : 0.6499929845800494

Interpretation

The AdaBoost model achieved a Mean Squared Error (MSE) of 0.000844075 and an R^2 score of 0.64999 on the test dataset.

Compared to the Random Forest model, AdaBoost produced a lower prediction error and explained a larger proportion of the observed variation in surface albedo. This suggests that the boosting strategy, which progressively focuses on observations that are more difficult to predict, helped the model capture additional environmental variability.

However, the baseline Decision Tree still achieved the strongest overall performance. The lower MSE and higher R^2 obtained by the Decision Tree indicate that the dominant relationships within the dataset were already represented effectively through a relatively simple set of environmental decision rules.

Overall, the AdaBoost results demonstrate that boosting can improve performance relative to some ensemble approaches, but additional model complexity did not translate into a performance gain beyond the baseline Decision Tree for this dataset. This finding highlights that model effectiveness depends not only on complexity, but also on how well the learning strategy matches the structure of the underlying data.

Comparing Model Performance

To evaluate the effectiveness of each machine learning approach, the test-set performance metrics are summarized below. Lower Mean Squared Error (MSE) values indicate smaller prediction errors, while higher R^2 values indicate a greater proportion of explained variability in Antarctic surface albedo.

```
In [14]: results=pd.DataFrame({"Model": ["Decision Tree","Random Forest","AdaBoost"],
                                "MSE": [mse_tree,mse_rf,mse_ada],
                                "RMSE": [np.sqrt(mse_tree),np.sqrt(mse_rf),np.sqrt(mse_ada)],
                                "R2": [r2_tree,r2_rf,r2_ada]})
                                results.sort_values("MSE")
```

Out[14]:

	Model	MSE	RMSE	R2
0	Decision Tree	0.000824	0.028708	0.658254
2	AdaBoost	0.000844	0.029053	0.649993
1	Random Forest	0.000926	0.030424	0.616185

Interpretation of Model Comparison

Among the three models evaluated, the *Decision Tree* produced the strongest results, achieving the lowest prediction error (MSE = 0.000824) and the highest R^2 score (0.658). This indicates that a relatively simple set of environmental decision rules was able to capture much of the observed variation in Antarctic surface albedo.

AdaBoost delivered the second-best performance, with results very close to those of the Decision Tree. By placing greater emphasis on observations that were harder to predict, it was able to capture additional environmental variability while maintaining a low prediction error.

The *Random Forest* model produced the lowest overall performance of the three approaches. Although ensemble averaging often improves stability, the averaging process may have smoothed some of the localized environmental patterns that were important for predicting albedo in this dataset.

Overall, all three models successfully identified meaningful relationships between the environmental variables and surface albedo. However, the Decision Tree provided the best combination of predictive accuracy, interpretability, and simplicity. The relatively small

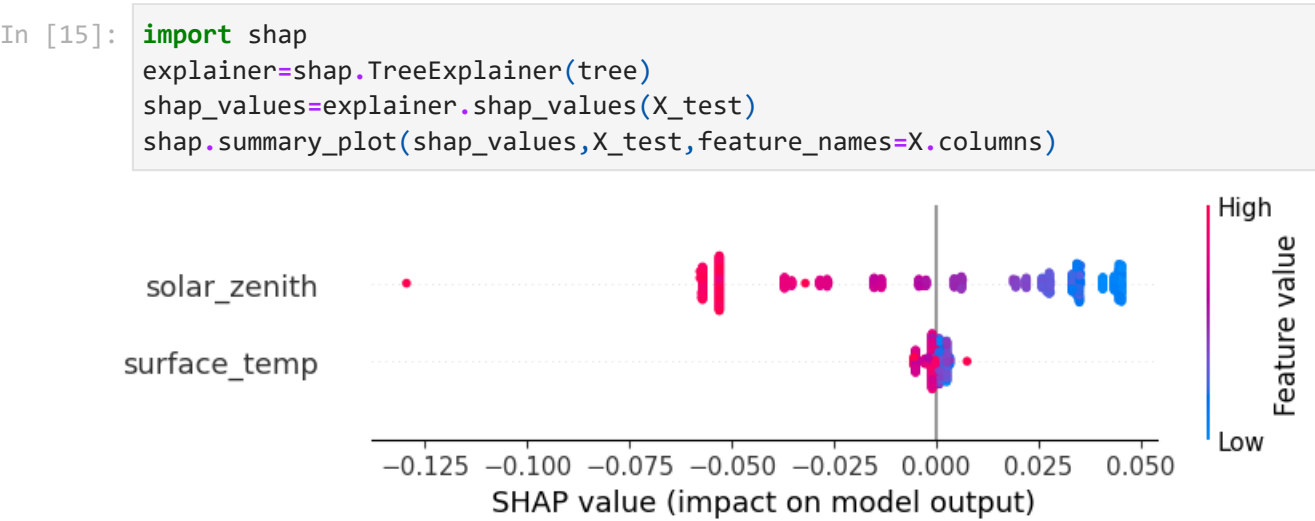
performance differences between the models also suggest that the dominant environmental signals within the dataset can be captured effectively without requiring highly complex modeling frameworks.

Model Explainability Using SHAP

Machine learning models can produce accurate predictions, but understanding how those predictions are generated is equally important. SHAP (SHapley Additive exPlanations) provides a way to quantify how each environmental variable contributes to a model's output.

The previous analyses showed that solar zenith angle consistently dominated the predictive process. SHAP allows us to investigate this relationship in detailed way by measuring how individual variables influence albedo predictions across all observations.

This helps connect machine learning results back to physical environmental processes and improves the interpretability of the model.



Interpretation

The SHAP summary plot ranks variables according to their contribution to model predictions. Consistent with the Decision Tree and Random Forest analyses, solar zenith angle emerged as the dominant predictor of Antarctic surface albedo, while surface temperature played a substantially smaller role.

The wide spread of SHAP values for solar zenith indicates that changes in illumination geometry can strongly influence predicted albedo across the Larsen C Ice Shelf. Lower solar zenith values generally contribute positively to predicted albedo, whereas higher solar zenith values contribute negatively.

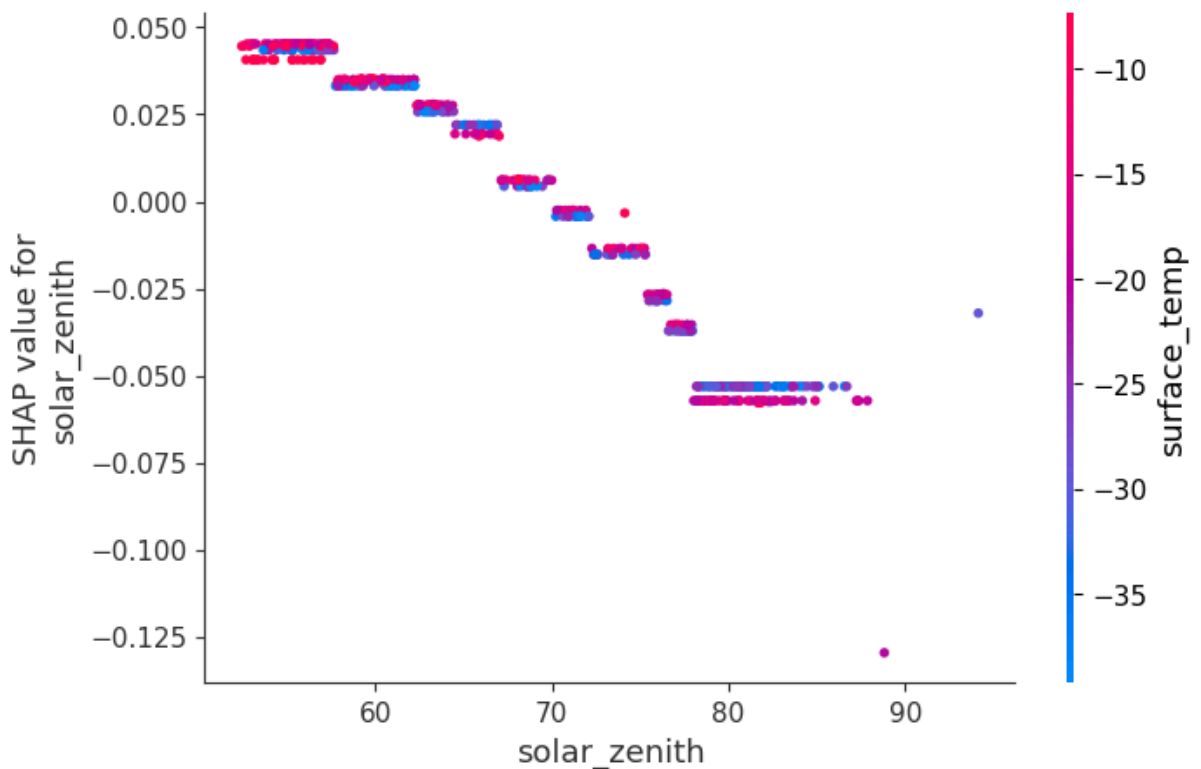
One notable feature of the plot is the presence of an extreme negative SHAP value associated with a high solar zenith observation. This isolated point indicates that under certain illumination conditions, solar geometry can put an unusually strong influence on the model's prediction, likely corresponding to rare low-albedo environmental states.

Surface temperature exhibits a much narrower SHAP range and remains tightly clustered around zero. This suggests that its influence is comparatively localized and mainly acts as a secondary adjustment factor rather than a primary driver of reflectance variability.

The SHAP analysis concludes that solar illumination geometry explains most of the observed variation in Antarctic surface albedo during the study period.

In [16]:

```
shap.dependence_plot("solar_zenith",shap_values,X_test,feature_names=X.columns)
```

Interpretation

The SHAP dependence plot illustrates how the influence of solar zenith angle changes across its observed range. As solar zenith increases, the SHAP contribution becomes progressively more negative, indicating a reduction in predicted surface albedo.

Instead of forming a smooth curve, the relationship appears as a staircase pattern. This behavior reflects the structure of tree-based models, which make predictions through a series of threshold-based splits rather than a continuous mathematical equation. Each step represents a decision boundary learned by the model.

The mixture of warm and cold temperature observations within each step suggests that temperature contributes only minor local adjustments, while the major changes in prediction are controlled by solar zenith thresholds.

A particularly notable feature appears at the highest solar zenith values, where the staircase drops sharply into a strongly negative SHAP region. This indicates that under extreme illumination conditions, solar geometry can put a much larger influence on the model's predictions and is closely associated with the lowest predicted albedo states identified in the dataset.

Therefore, the SHAP results show that the model uses solar angles to perform most of the predictive work when estimating Antarctic surface reflectivity, while temperature primarily provides smaller refinements within those broader illumination regimes.

Conclusion

This two-part study investigated whether machine learning techniques could model Antarctic surface albedo dynamics using a reduced set of satellite-derived environmental indicators extracted over the Larsen C Ice Shelf between 2015 and 2025.

Part 1 established a baseline using a custom Degree-2 Polynomial Regression model optimized through Gradient Descent. The model successfully captured the dominant reflectance behavior of the ice shelf and demonstrated that meaningful environmental relationships could be extracted from a lightweight tabular representation of Earth Observation data. However,

diagnostic analysis revealed reduced performance under uncommon low-albedo conditions, indicating limitations associated with a single global regression surface.

Part 2 extended the analysis using Decision Tree, Random Forest, and AdaBoost regression models. Among the models tested, the Decision Tree produced the achieved strong overall performance, having the lowest prediction error and highest R^2 score. This suggests that the environmental relationships present within the dataset were represented effectively through a relatively simple set of decision rules.

The explainability results showed a clear and consistent pattern across the different models. Solar zenith angle was identified as the primary variable influencing surface albedo, while surface temperature mainly helped refine predictions under specific conditions. Similar conclusions were obtained from the Decision Tree, Random Forest, and SHAP analyses.

Overall, this study shows that machine learning can be used to model Antarctic surface albedo using only a small number of environmental variables. The results consistently identified solar zenith angle as the most important factor influencing albedo across the Larsen C Ice Shelf. While there is still scope to improve predictions by including additional variables and higher-resolution data, these findings demonstrate that even relatively simple models can provide useful insights into the environmental processes that drive Antarctic surface reflectivity.
